

Nyenje Bay Wind/Waves Analysis Tool

Technical Framework Report

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Location: **Nyenje Bay, Lake Kariba**

Coordinates: **16.53S, 28.83E**

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Executive Summary

This report documents the technical framework of the Nyenje Bay Wind/Waves Analysis Tool. The tool integrates multiple layers:

- **User Interface Layer:** Streamlit-based web dashboard
- **Scientific Computing Layer:** NumPy, SciPy, Pandas, Xarray
- **Data Layer:** ERA5 reanalysis, CCMP satellite, ground measurements
- **Wave Modeling Layer:** SWAN model integration

Key Scientific Calculations:

- Wind averaging periods: 3-minute, 10-minute, 3-second gust, 10-second gust
- Gust factors: 1.15 (3-min), 1.45 (3-sec), 1.35 (10-sec)
- JONSWAP wave height formula
- Gumbel extreme value analysis for return periods

1 Tool Architecture

1.1 Four-Layer Framework

The tool is built on a four-layer architecture:

Layer	Technology	Functions
User Interface	Streamlit	Web dashboard, interactive plots
Scientific Computing	NumPy, SciPy	Wind averaging, wave calculations
Data Processing	Xarray, Pandas	ERA5 reading, grid interpolation
External Models	SWAN	Wave distribution

Table 1: Architecture Layers

2 Wind Averaging Calculations

2.1 Overview

The tool calculates five different wind averaging periods:

Period	Gust Factor	Application
Hourly Mean	1.00	Reference wind speed
3-Minute Average	1.15	Dynamic response
10-Minute Average	1.00	Marine forecast
3-Second Gust	1.45	Ultimate Limit State (ULS)
10-Second Gust	1.35	Serviceability Limit State (SLS)

Table 2: Wind Averaging Periods

2.2 Exponential Weighted Moving Average

For hourly data, the tool uses Exponential Weighted Moving Average:

$$\text{EWMA}_t = \alpha \cdot x_t + (1 - \alpha) \cdot \text{EWMA}_{t-1} \quad (1)$$

Where:

- $\alpha = \frac{2}{\text{span}+1}$ is the smoothing factor
- For 3-minute average: $\text{span} = 3$, $\alpha = 0.5$
- For 10-minute average: $\text{span} = 10$, $\alpha \approx 0.18$

2.3 Gust Factor Formula

$$\text{Gust Factor} = \frac{\text{Peak Gust Speed}}{\text{Mean Wind Speed}} \quad (2)$$

Period	Gust Factor
3-second gust	1.45
10-second gust	1.35
3-minute average	1.15

Table 3: Gust Factor Derivation

3 Wave Height Calculation

3.1 JONSWAP Formula

The Joint North Sea Wave Project (JONSWAP) formula is used for fetch-limited waves.

3.1.1 Dimensionless Fetch

$$\tilde{x} = \frac{g \cdot F}{U^2} \quad (3)$$

Where:

- $g = 9.81 \text{ m/s}^2$ (gravity)
- $F = \text{fetch length (meters)}$
- $U = \text{wind speed (m/s)}$

3.1.2 Significant Wave Height

$$H_s = 0.0016 \cdot \frac{U^2}{g} \cdot \sqrt{\tilde{x}} \quad (4)$$

3.1.3 Peak Wave Period

$$T_p = 0.286 \cdot \frac{U}{g} \cdot (\tilde{x})^{0.33} \quad (5)$$

3.2 Simplified Formula

$$H_s = 0.0016 \cdot U \cdot \sqrt{g \cdot F} \quad (6)$$

3.3 Example Calculations

Wind Speed (m/s)	Wave Height H_s (m)	Period T_p (s)
5	0.07	1.2
8	0.18	2.0
10	0.29	2.3
12	0.41	2.7
15	0.64	3.2
18	0.92	3.7
20	1.14	4.0

Table 4: Wave Height for Different Wind Speeds (Fetch = 4 km)

4 Extreme Value Analysis

4.1 Gumbel Distribution

The cumulative distribution function (CDF) is:

$$F(x) = \exp \left(- \exp \left(- \frac{x - \mu}{\beta} \right) \right) \quad (7)$$

Where:

- μ = location parameter
- β = scale parameter

4.2 Return Period Calculation

For return period T (years):

$$x(T) = \mu - \beta \cdot \ln \left(- \ln \left(1 - \frac{1}{T} \right) \right) \quad (8)$$

4.3 Results for Nyenje Bay

Return Period (years)	Wind Speed (m/s)	Wave Height (m)
2	11.2	0.85
5	12.8	1.02
10	14.0	1.15
20	15.2	1.28
25	15.6	1.32
50	16.8	1.48
100	18.0	1.62

Table 5: Return Period Wave Heights

5 Validation Results

5.1 Kariba Airport Comparison

Metric	Raw ERA5	Debiased ERA5	Improvement
Correlation (r)	0.39	0.502	+29%
Bias (m/s)	+2.04	+0.01	99.5%
RMSE (m/s)	2.45	0.52	79%

Table 6: Validation Statistics (2001-2005)

5.2 Interpretation of $r = 0.502$

$$r = 0.502 \quad \Rightarrow \quad R^2 = 0.252 \quad (9)$$

This means:

- 25.2% of wind variance is shared between locations
- 74.8% is due to local effects (expected for 3 km separation)
- The correlation exceeds the expected value of 0.40

6 Technology Stack

Component	Technology	Purpose
UI Framework	Streamlit	Web dashboard
Scientific	NumPy	Array operations
Data Analysis	Pandas	DataFrames
NetCDF	Xarray	ERA5 reading
Statistics	SciPy	Gumbel distribution
Visualization	Plotly	Interactive plots
Wave Model	SWAN	Wave distribution

Table 7: Technology Stack

7 Conclusions

7.1 Key Technical Achievements

1. **Multi-Layer Architecture:** Separation of UI, computation, and data
2. **Wind Averaging:** EWMA method for 3-min and 10-min averages
3. **Gust Factors:** Engineering-standard factors (1.45, 1.35)
4. **Wave Calculations:** JONSWAP fetch-limited formula
5. **Extreme Analysis:** Gumbel distribution for return periods
6. **Validation:** $r = 0.502$ correlation with ground measurements

7.2 Framework Independence

The scientific core is completely independent of the user interface:

- Same calculations work in command line, web app, or API
- Streamlit is only for visualization, not computation
- Core functions can be reused in other projects